CARLETON UNIVERSITY

AERO 4402 A

Aerospace Propulsion

CFD Assignment

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1.0 Introduction

This Computational Fluid Dynamics (CFD) assignment focuses on the aerodynamic analysis of a turbofan engine's fan stage, using a combination of MDIDS-GT, ANSYS Turbo-Grid, and ANSYS CFX. The project objective is to iteratively design and evaluate fan air-foil geometries with the aim of minimizing flow separation, reducing total pressure loss, and achieving a mass flow rate consistent with MDIDS predictions.

The initial phase of the project involved the geometric design of the fan stage in MDIDS-GT, where key air-foil components—including the stator, rotor, strut, and bypass strut—were individually modelled. The generated 2D mean-line profiles were then extended into 3D blade geometries, respecting aerodynamic design parameters such as stagger angle, leading edge (LE) and trailing edge (TE) diameters, and maximum thickness.

Once the air-foil geometries were defined, they were exported to ANSYS Turbo-Grid for structured mesh generation. Following successful meshing, the CFD simulations were performed in ANSYS CFX. The analysis was guided by the MDIDS mean-line results, which were used as inlet and outlet boundary conditions. These served as reference points to evaluate how well the CFD-predicted mass flow rate matched the MDIDS predictions within the allowable ±1–3% tolerance.

Three iterative design and simulation cycles were completed during the course of this assignment. Each iteration involved targeted modifications to air-foil parameters—particularly those affecting the stator, rotor, strut, and bypass strut. Parameters adjusted included stagger angles, LE/TE diameters, and blade thickness profiles, with the aim of improving aerodynamic performance and mass flow convergence.

The report that follows outlines the evolution of the air-foil designs, simulation methodology, analysis of results, and lessons learned throughout this iterative CFD design process.

2.0 MDIDS, Geometry and Meshing

All air-foil components (rotor, stator, strut, and bypass stator) were initially designed in MDIDS-GT. Geometries were exported to ANSYS Turbo-Grid to generate structured meshes. Minor meshing artifacts, such as sharp edges, were noted, potentially affecting convergence quality.

Meshing was consistent across simulations to isolate changes from design parameters rather than mesh variations.

MDIDS Section

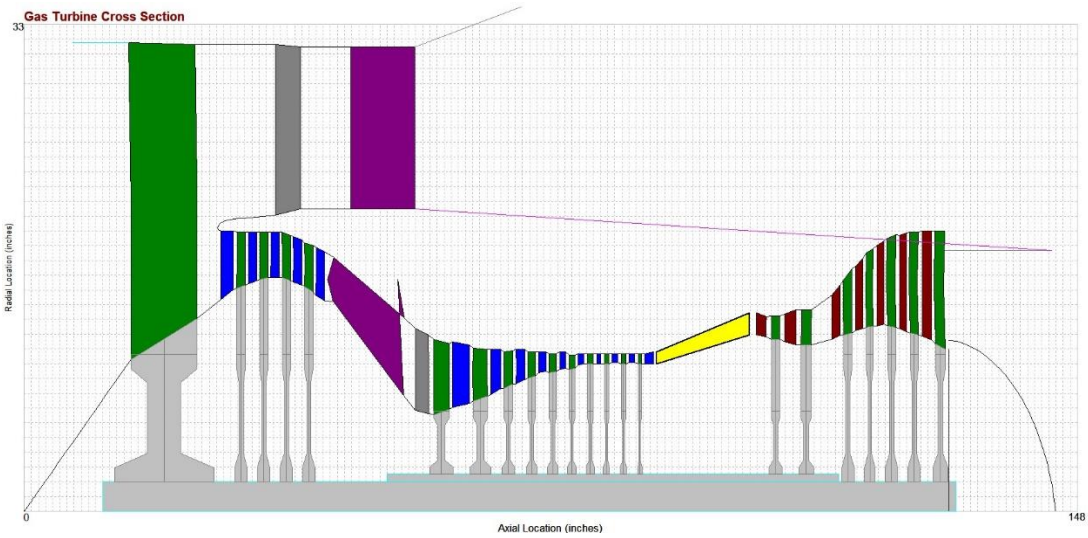

Figure 1: The Complete MDIDS gas turbine cross section MDIDS

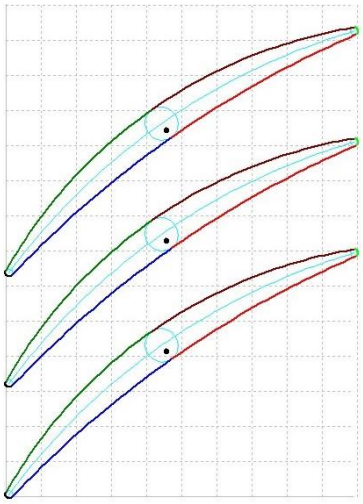

Figure 2: Rotor 2D airfoil single section

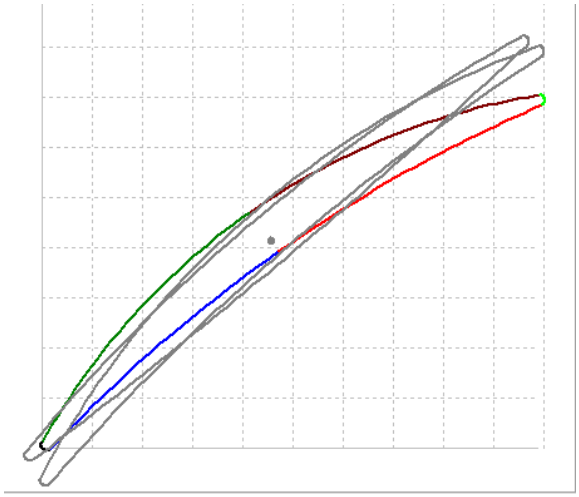

Figure 3: Rotor 2D air-foil all sections

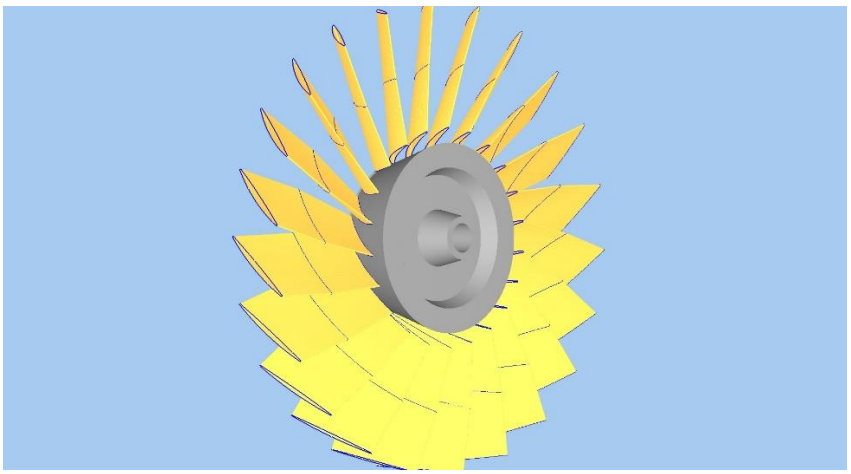

Figure 4: Rotor 3D

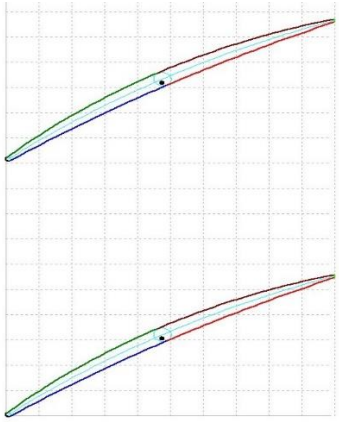

Figure 5: Stator 2D air-foil single section

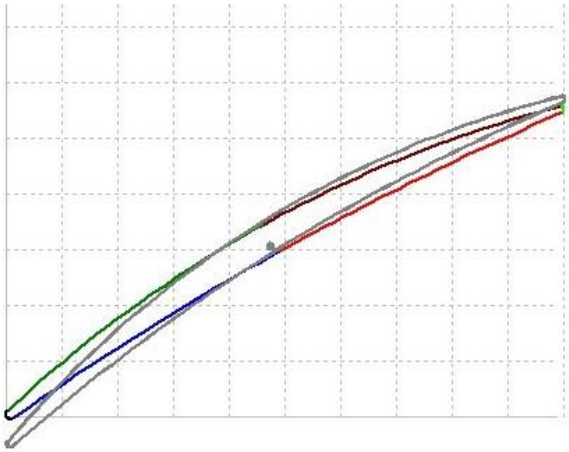

Figure 6: Stator 2D air-foil all sections

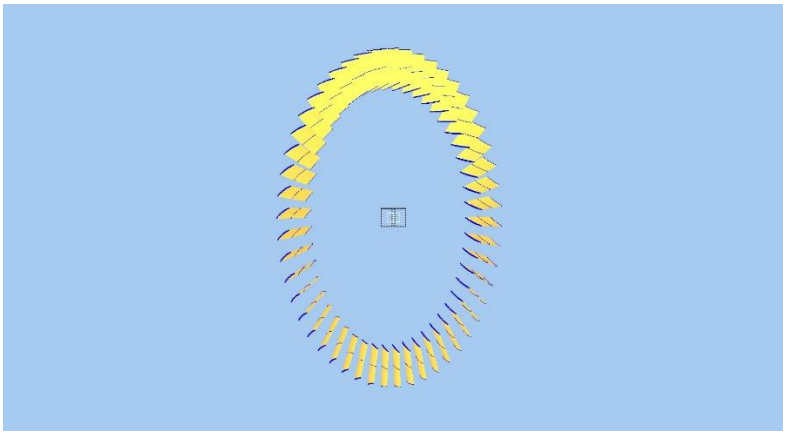

Figure 7:Stator 3D

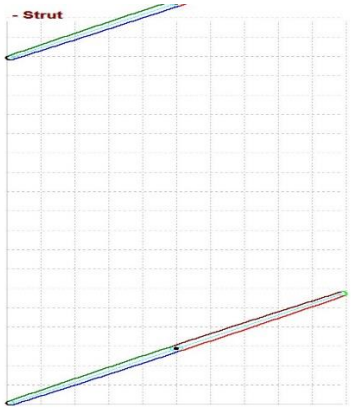

Figure 8: Strut 2D air-foil single section

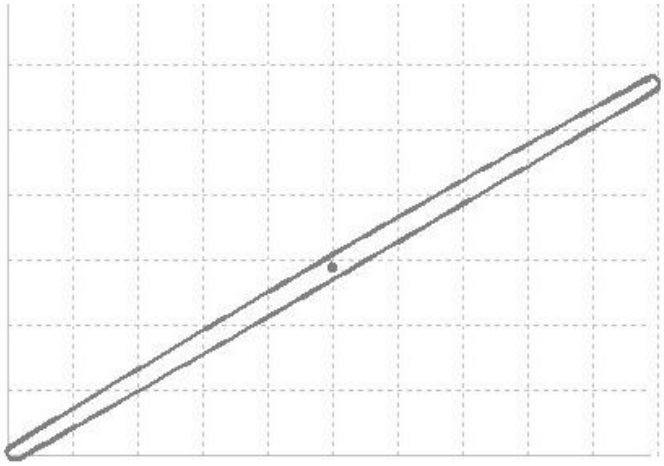

Figure 9: Strut 2D air-foil all sections

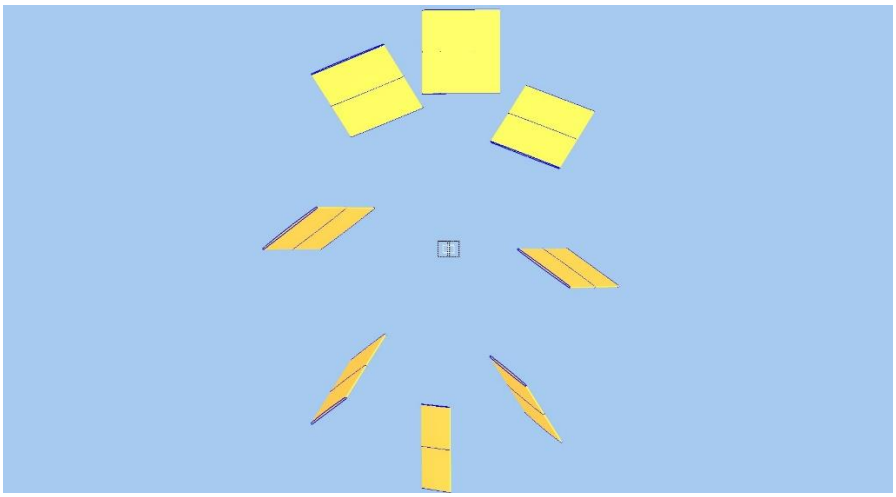

Figure 10: Strut 3D

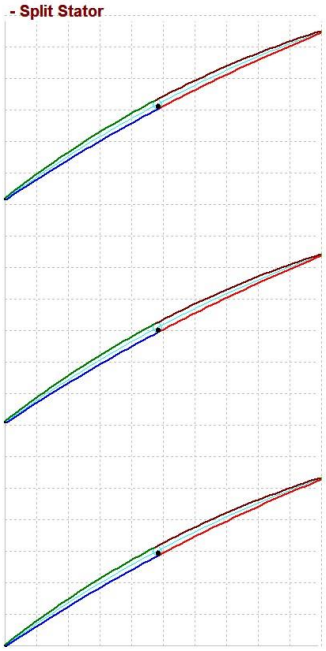

Figure 11: By-pass stator 2D air-foil single section

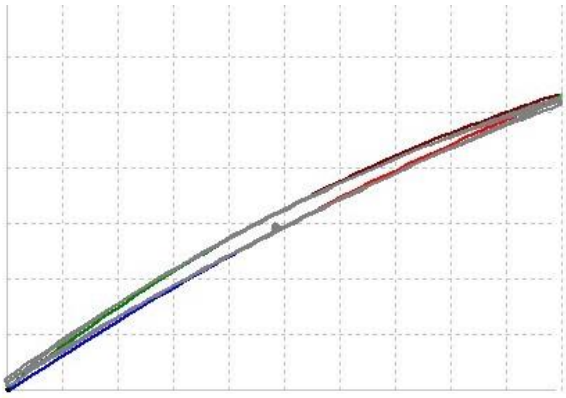

Figure 12: By-pass stator 2D air-foil All sections

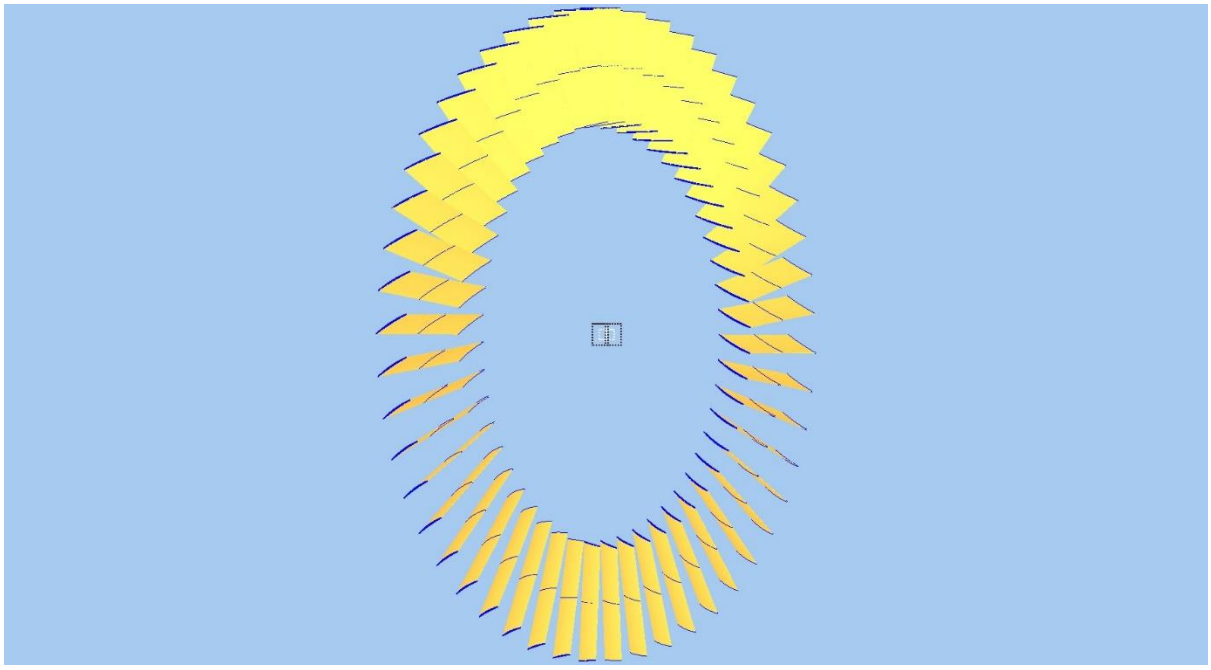

Figure 13: By-pass stator 3D

ANSYS

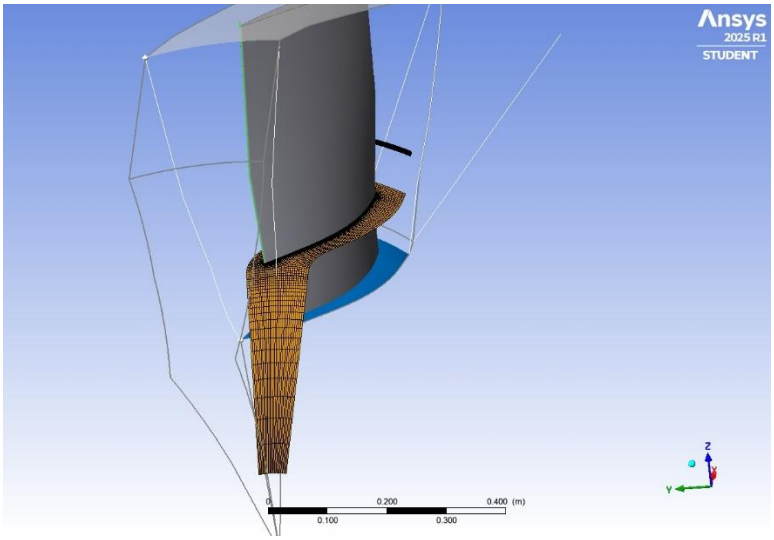

Figure 14: Meshed Fan rotor

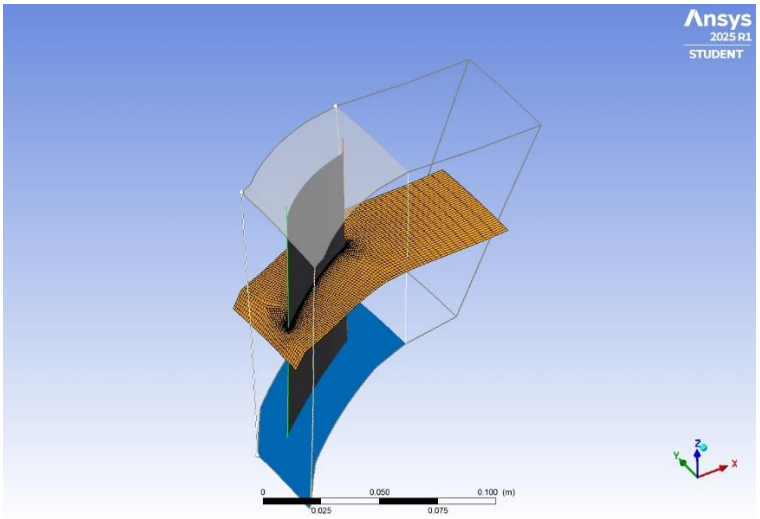

Figure 15: Meshed Stator

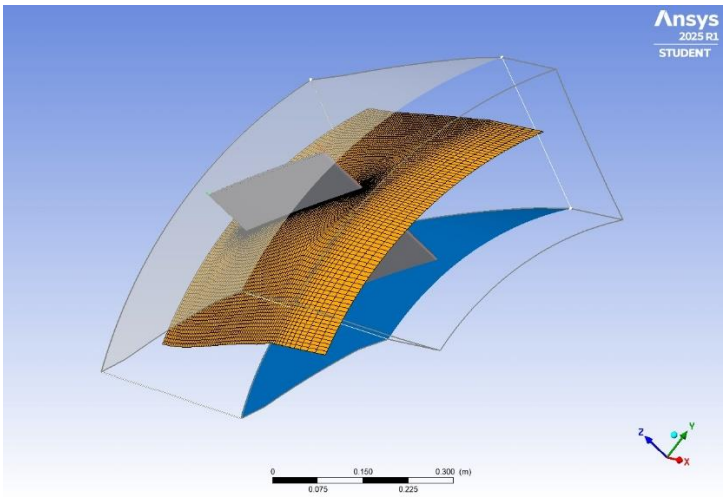

Figure 16: Meshed Strut

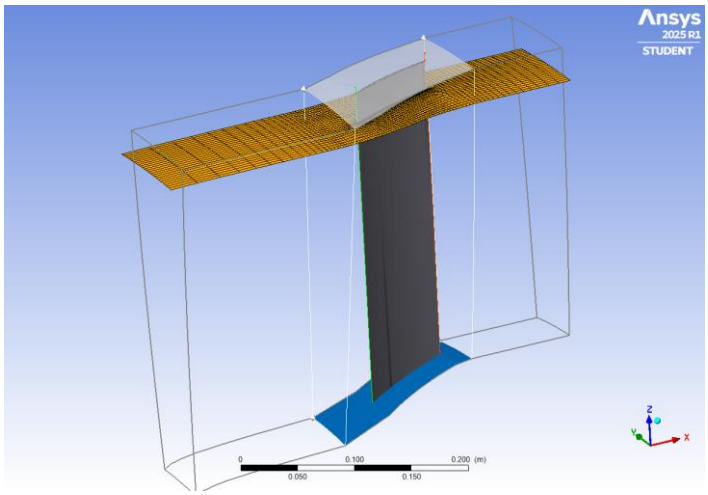

Figure 17: Meshed By-pass stator

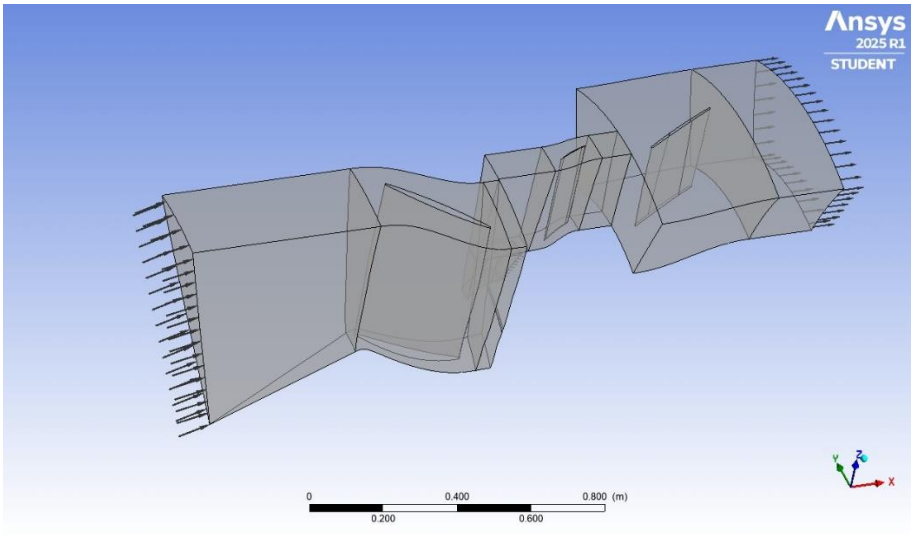

Figure 18: Meshed fan stage

The rotor, stator, strut, and bypass stator were all meshed in ANSYS Turbo-Grid, as shown in Figures 13 to 16. These meshes were used consistently across all simulation iterations.

Methodology

The CFD analysis followed a structured three-iteration approach. The geometry for the rotor, stator, strut, and bypass stator was initially designed in MDIDS-GT. Key aerodynamic parameters—stagger angle, leading edge (LE) diameter, trailing edge (TE) metal angle, and maximum thickness—were adjusted in each iteration to optimize flow characteristics and reduce performance losses.

The first two iterations were run for 100 iterations each in ANSYS CFX, which provided an early performance baseline. For the final iteration, 500 iterations were conducted to improve solution convergence and flow stability. In addition to geometry changes, boundary conditions such as AF2 and AF3 outlet pressures were modified between iterations to bring the mass flow rate closer to MDIDS predictions.

Results

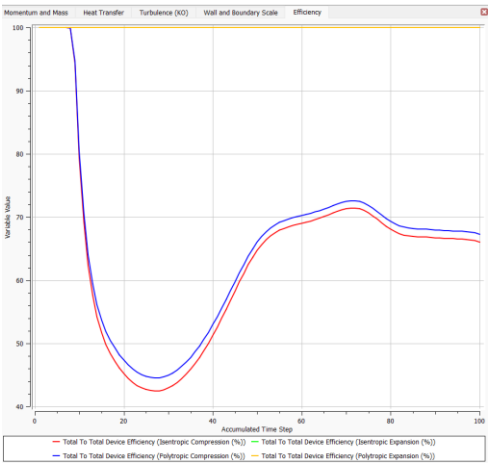

Figure 19: Efficiency plot for iteration 1

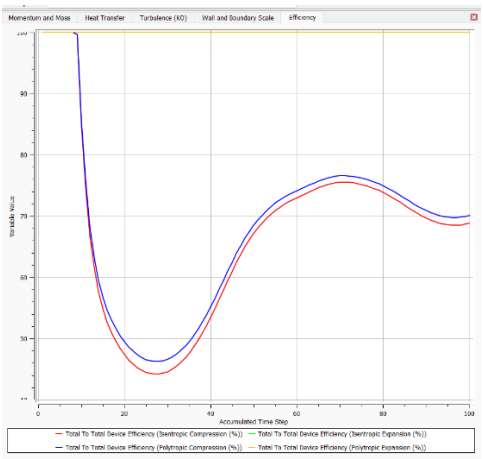

Figure 20: Efficiency plot for iteration 2.

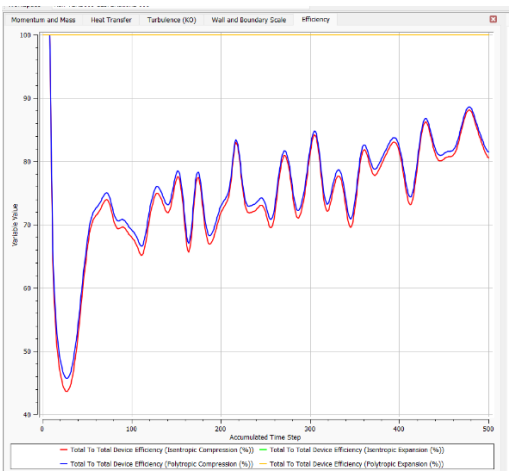

Figure 21: Efficiency plot for iteration 3

The efficiency plots from all three iterations illustrate how solver stability and aerodynamic performance improved over time. In the first iteration, the efficiency dropped sharply before gradually recovering and stabilizing around 67% (polytropic) and 65% (isentropic), but it flattened early, indicating limited convergence due to the low iteration count. The second iteration followed a

similar pattern, with slightly improved values (~70% and ~68%, respectively), suggesting minor geometry and boundary condition changes had a positive but limited effect. The third iteration, run with 500 iterations, showed significantly better performance. Although the plot displayed oscillations, the overall trend rose steadily, stabilizing near 87% (polytropic) and 85% (isentropic). This iteration demonstrated that increased iteration count allowed the solver to more accurately capture flow behaviour and reach a deeper, more stable solution. Overall, the efficiency plots confirmed that both input adjustments and sufficient solver runtime are essential for achieving accurate and optimized results.

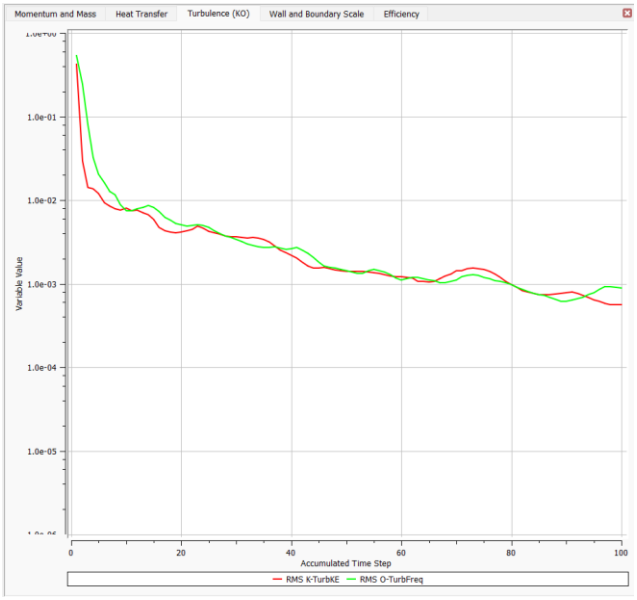

Figure 22: Turbulence plot for iteration 1

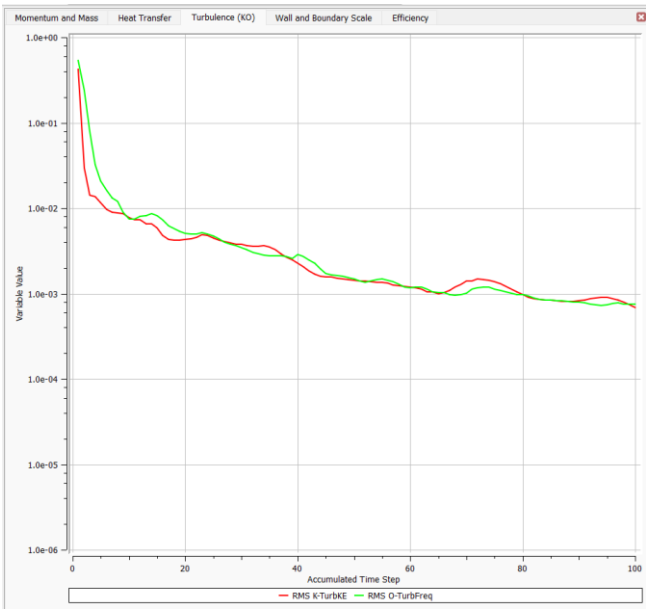

Figure 23: Turbulence plot for iteration 2

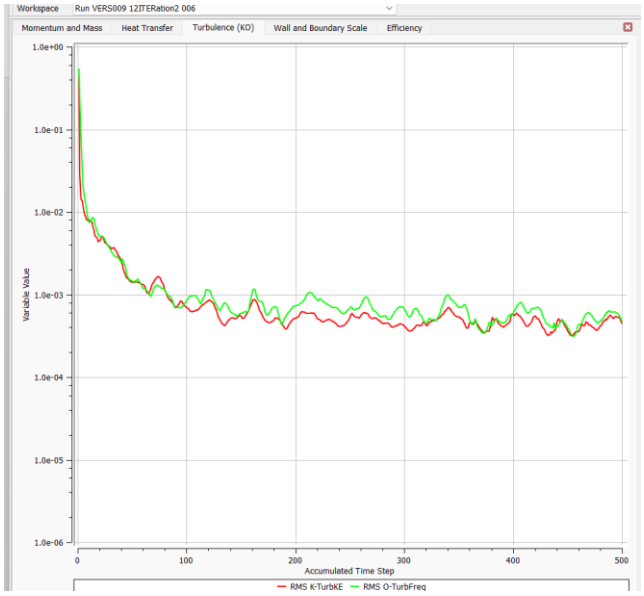

Figure 24: Turbulence plot for iteration 3

The turbulence plots (RMS k-TurbKE and ω-TurbFreq) show a consistent trend of declining values across all three iterations, indicating improved stability with each successive design. In Iterations 1 and 2, both variables dropped steadily over 100 steps, levelling out near 1e-03 with some minor oscillations. However, in Iteration 3, which ran for 500 steps, the solution reached lower turbulence residuals earlier and maintained a tighter, more stable band—despite small fluctuations after 200 steps. This behaviour suggests that the final design, combined with the higher iteration count, allowed for better turbulence model convergence and a more resolved flow field

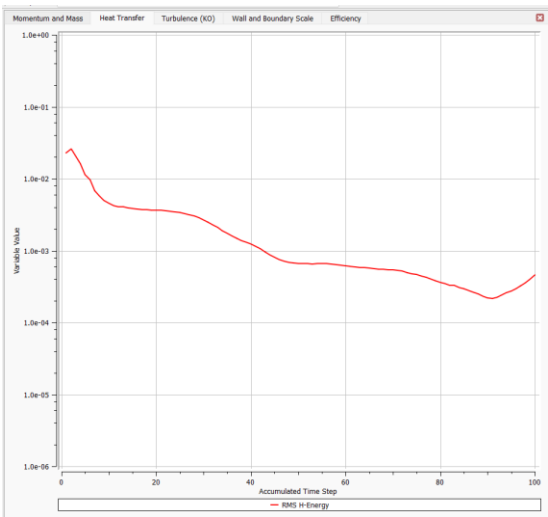

Figure 25: Heat transfer plot for iteration 1

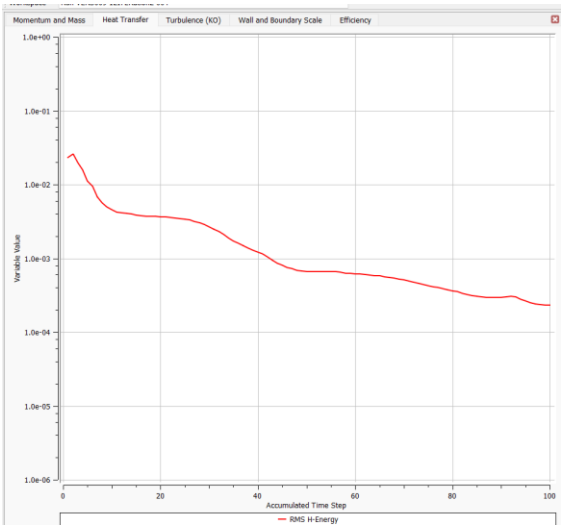

Figure 26: Heat transfer plot for iteration 2

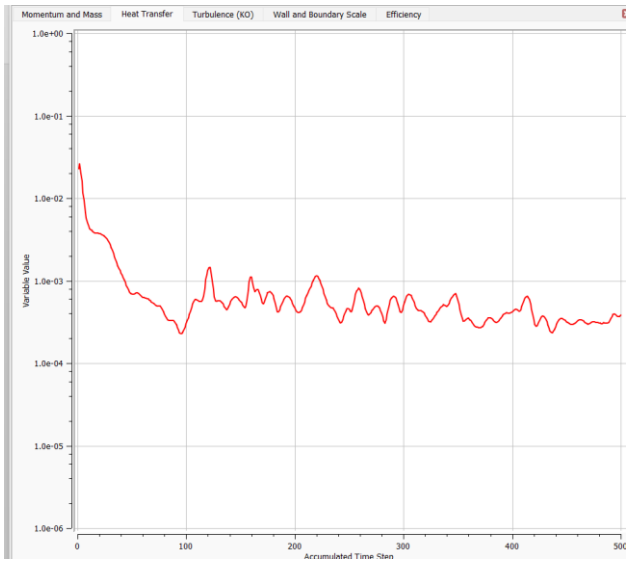

Figure 27: Heat transfer plot for iteration 3

The heat transfer convergence plots show a clear improvement in stability and residual reduction across iterations. In Iterations 1 and 2 (100 steps), the RMS H-Energy steadily declines from ~1e-02 to below 1e-04, with a smoother curve in the second iteration indicating slightly better energy resolution. Iteration 3, which ran for 500 iterations, initially followed a similar drop but then fluctuated around 1e-04. Although the energy residuals in the final iteration didn't settle as smoothly, the longer run allowed the solution to reach a more refined state, capturing the flow energy dynamics more accurately. The slight oscillations are expected in more complex, higher-resolution simulations.

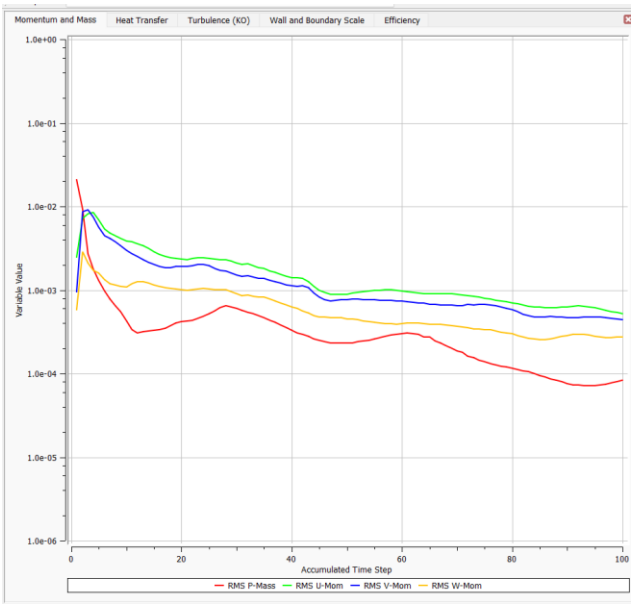

Figure 28: Momentum and Mass plot for iteration 1

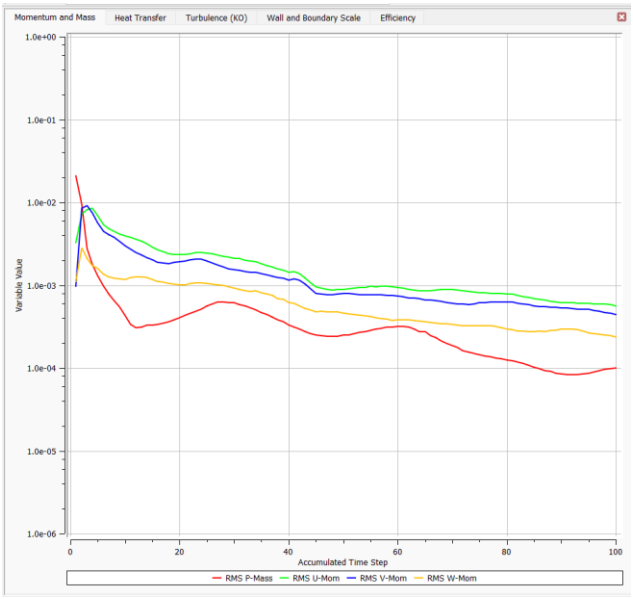

Figure 29: Momentum and Mass for iteration 2

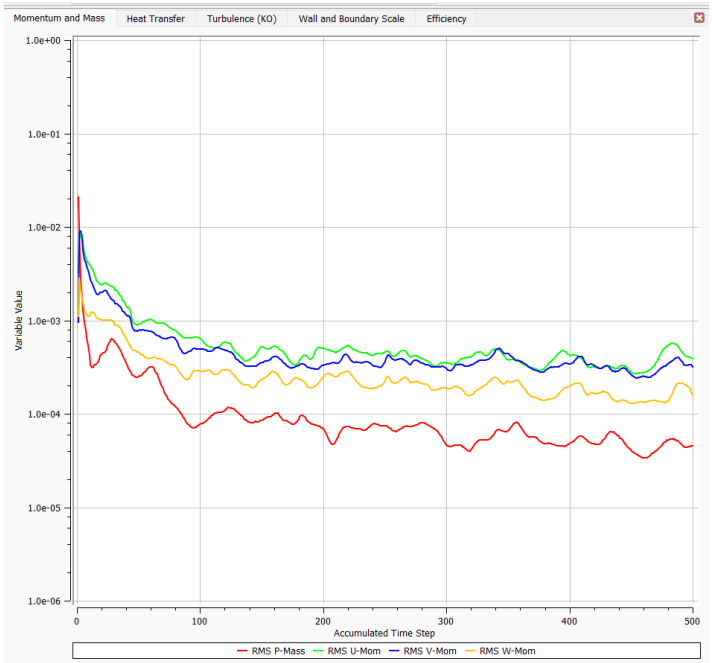

Figure 30: Momentum and mass plot for iteration 3

The mass and momentum plots across the three iterations reflect the solver's ability to stabilize residuals over time. In Iterations 1 and 2 (100 steps), RMS values for mass (P-Mass) and momentum components (U, V, W) consistently decreased, with residuals trending below 1e-03 by the end. However, minor flattening and slight variability remained, indicating that full convergence hadn't been achieved. In contrast, Iteration 3 (500 steps) displayed lower residuals overall, particularly for mass and V/W momentum, though mild fluctuations persisted throughout the run. These results suggest that the extended iteration count allowed for better residual damping and a more balanced momentum field across the domain.

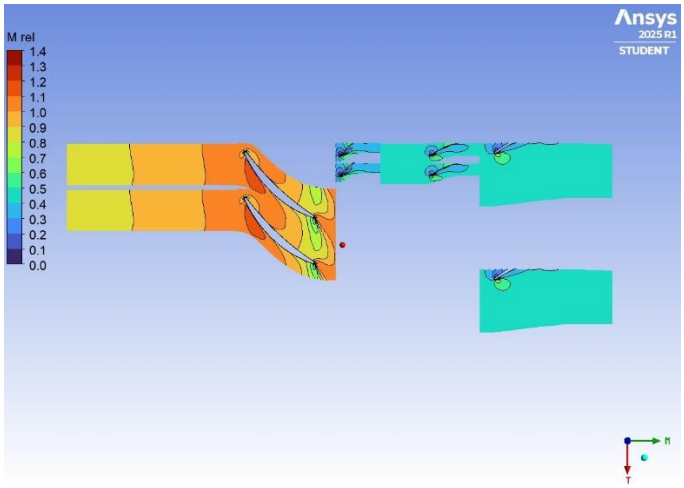

Figure 31: Mach number at 50% span for iteration 1

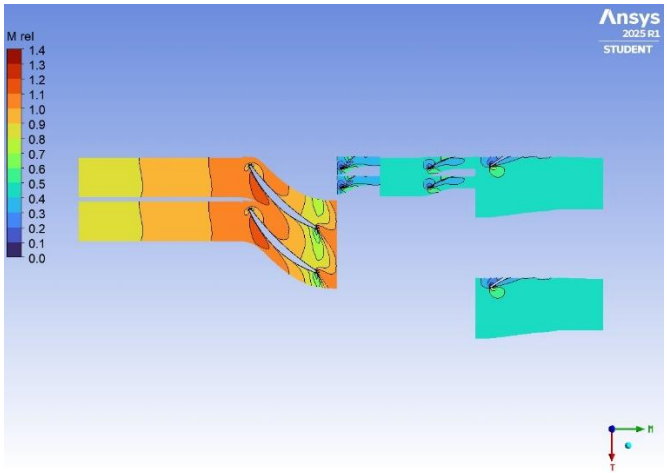

Figure 32: Mach number at 50% span for iteration 2

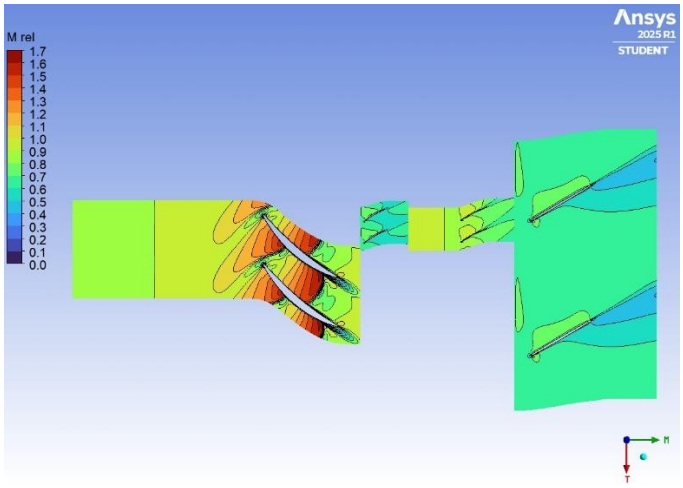

Figure 33: Mach number at 50% span for iteration 3

The Mach number contours across all three iterations show a clear progression in flow quality and separation control. In Iterations 1 and 2, low-Mach blue zones around the trailing edges indicate moderate flow separation, especially near the bypass stator and rotor tips. These regions exhibit disrupted streamlines and stagnant zones. By Iteration 3, the flow field becomes significantly cleaner—streamlines remain attached, and separation is minimized. The Mach range also broadens slightly, showing more efficient acceleration through the blades. This improvement highlights the effectiveness of refined geometry and outlet pressure tuning in controlling adverse pressure gradients.

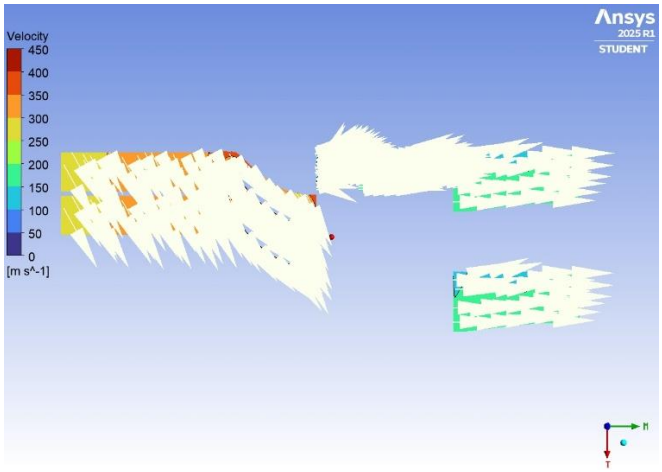

Figure 34: Velocity vectors at 50% span for iteration 1

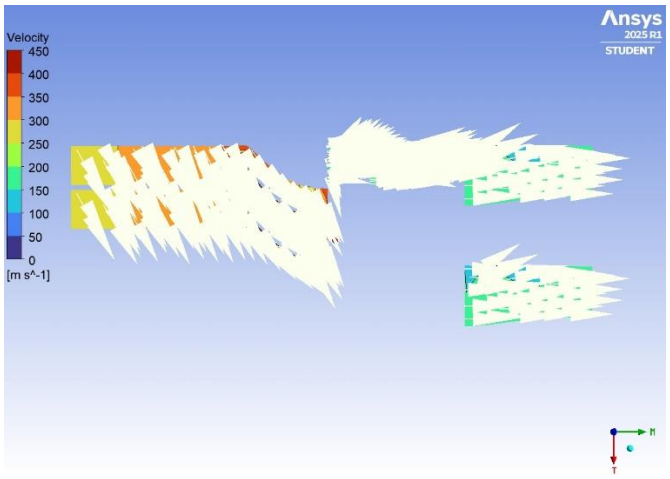

Figure 35: Velocity vectors at 50% span for iteration 2

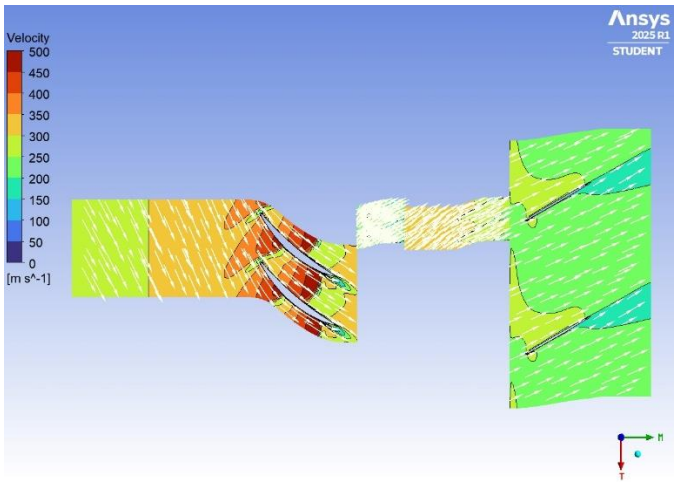

Figure 36: Velocity vectors at 50% span for iteration 3

Table 1: Overall Performance for iteration 1

Table 1. Overall Performance Results Table

Inlet Mass Flow Rate	322.4050	[kg s^-1]
Inlet Volume Flow Rate	260.1030	[m^3 s^-1]
Total Pressure Ratio	1.2646	
Total Temperature Ratio	1.0282	
Total-to-Total Isentropic Efficiency %	100.0000	
Total-to-Total Polytropic Efficiency %	100.0000	

Table 2: Overall Performance for iteration 2

Table 1. Overall Performance Results Table

Inlet Mass Flow Rate	322.4050	[kg s^-1]
Inlet Volume Flow Rate	260.1030	[m^3 s^-1]
Total Pressure Ratio	1.2646	
Total Temperature Ratio	1.0282	
Total-to-Total Isentropic Efficiency %	100.0000	
Total-to-Total Polytropic Efficiency %	100.0000	

Table 3: Overall Performance for iteration 3

Table 1. Overall Performance Results Table

Inlet Mass Flow Rate	280.5230	[kg s^-1]
Inlet Volume Flow Rate	229.8020	[m^3 s^-1]
Total Pressure Ratio	1.4112	
Total Temperature Ratio	1.2322	
Total-to-Total Isentropic Efficiency %	53.5239	
Total-to-Total Polytropic Efficiency %	55.7282	

The overall performance metrics show distinct trends across the three iterations when compared to the MDIDS targets of 258.53 kg/s mass flow, a total pressure ratio of 1.385, and 79.07% isentropic efficiency. Iterations 1 and 2 yielded nearly identical values, with a mass flow of 322.4 kg/s, a pressure ratio of 1.2646, and an unrealistic 100% efficiency—indicating incomplete convergence or incorrect reporting from CFX. Iteration 3, however, produced more realistic values: a mass flow rate of 280.5 kg/s, pressure ratio of 1.4112, and an isentropic efficiency of 53.5%. While the mass flow and pressure ratio were closer to MDIDS predictions, the lower efficiency indicates remaining flow losses and a need for further refinement. Still, Iteration 3 showed the best trade-off between realism and accuracy.

Table 4: Stage 1 performance results for iteration 1

Table 2. Stage 1 Performance Results

Rotation Speed	-591.6670	[radian s^-1]
Inlet Mass Flow Rate	322.4050	[kg s^-1]
Inlet Volume Flow Rate	260.1030	[m^3 s^-1]
Reference Radius	0.5914	[m]
Input Power	2673770.0000	[W]
Inlet Flow Coefficient	0.4738	
Total Pressure Ratio	1.2646	
Total Temperature Ratio	1.0282	
Polytropic Head	20057.7000	[J kg^-1]
Polytropic Head Coefficient	0.1583	
Total-to-Total Polytropic Efficiency %	241.8570	
Diffuser Loss Coef., Y1	0.1688	
Diffuser Loss Coef., Y2	0.2031	
Diffuser Loss Coef., Y3	0.6322	
Diffuser Loss Coef., Y4	1.4906	
Diffuser Cp	-0.0564	

Table 5: Stage 1 performance results for iteration 2

Table 2. Stage 1 Performance Results

Rotation Speed	-591.6670	[radian s^-1]
Inlet Mass Flow Rate	322.4050	[kg s^-1]
Inlet Volume Flow Rate	260.1030	[m^3 s^-1]
Reference Radius	0.5914	[m]
Input Power	2673770.0000	[W]
Inlet Flow Coefficient	0.4738	
Total Pressure Ratio	1.2646	
Total Temperature Ratio	1.0282	
Polytropic Head	20057.7000	[J kg^-1]
Polytropic Head Coefficient	0.1583	
Total-to-Total Polytropic Efficiency %	241.8570	
Diffuser Loss Coef., Y1	0.1688	
Diffuser Loss Coef., Y2	0.2031	
Diffuser Loss Coef., Y3	0.6322	
Diffuser Loss Coef., Y4	1.4906	
Diffuser Cp	-0.0564	

Table 6: Stage 1 performance results for iteration 3

Table 2. Stage 1 Performance Results

Rotation Speed	-591.6670	[radian s^-1]
Inlet Mass Flow Rate	280.5230	[kg s^-1]
Inlet Volume Flow Rate	229.8020	[m^3 s^-1]
Reference Radius	0.5911	[m]
Input Power	18853800.0000	[W]
Inlet Flow Coefficient	0.4826	
Total Pressure Ratio	1.4112	
Total Temperature Ratio	1.2322	
Polytropic Head	35057.8000	[J kg^-1]
Polytropic Head Coefficient	0.2771	
Total-to-Total Polytropic Efficiency %	52.1621	
Diffuser Loss Coef., Y1	0.1863	
Diffuser Loss Coef., Y2	0.2290	
Diffuser Loss Coef., Y3	0.4122	
Diffuser Loss Coef., Y4	1.2746	
Diffuser Cp	0.2644	

The Stage 1 performance results reveal critical differences between iterations, particularly in poly-tropic head, efficiency, and diffuser losses. Iterations 1 and 2 produced identical values with inflated poly-tropic efficiency (241.86%) and constant input power, suggesting non-physical outputs due to insufficient solver convergence. In contrast, Iteration 3 displayed more realistic and consistent results: poly-tropic efficiency dropped to 52.16%, with increased input power (18.85 MW) and improved pressure and temperature ratios. The poly-tropic head more than doubled, indicating a stronger energy transfer to the airflow. Additionally, diffuser loss coefficients dropped slightly, and the positive diffuser Cp in Iteration 3 implies improved recovery. These values collectively point to a more physically accurate and better-resolved simulation in the final iteration.

Table 7: Component 1 summary for iteration 1

Table 3. Component 1 Summary Data Table

Quantity	Inlet	LE Cut	TE Cut	Outlet	TE/LE	TE-LE	Units
Density	1.1613	1.2421	1.2695	1.3790	1.0221	N/A	[kg m^-3]
Pstatic	94982.9000	102592.0000	116043.0000	116297.0000	1.1311	13450.9000	[Pa]
Pt	104057.0000	121912.0000	158324.0000	132903.0000	1.2987	36412.5000	[Pa]
T	284.7140	284.9560	317.2800	293.4290	1.1134	32.3233	[K]
Tt	292.3910	300.1100	348.1220	305.3450	1.1600	48.0119	[K]
H	-13495.3000	-13251.7000	19213.9000	-4742.0300	-1.4499	32465.5000	[J kg^-1]
Ht	-5784.4300	1969.0800	50192.3000	7226.2800	25.4902	48223.2000	[J kg^-1]
Ht adiabatic	-5783.5400	5579.3200	38570.4000	15311.7000	6.9131	32991.0000	[J kg^-1]
Ht polytropic	-5784.4500	5491.4600	38479.8000	15010.4000	7.0072	32988.3000	[J kg^-1]
Rothalpy	-5783.6500	2353.9200	-60.5827	-243.0550	-0.0257	-2414.5100	[J kg^-1]
Entropy	-26.9859	-38.9235	9.1678	-53.0549	-0.2355	48.0913	[J kg^-1 K^-1]
Mach (abs)	0.3619	0.4991	0.6519	0.4384	1.3062	0.1528	
Mach (rel)	1.0221	1.1257	0.7991	1.0801	0.7099	-0.3266	
U	318.1090	343.9490	355.9150	359.6400	1.0348	11.9654	[m s^-1]
Cm	122.4190	165.8090	181.7260	148.8920	1.0960	15.9172	[m s^-1]
Cu	0.0422	1.9752	140.7510	20.7577	71.2572	138.7760	[m s^-1]
C	122.4200	169.0640	233.6020	150.5600	1.3817	64.5381	[m s^-1]
Wu	-318.0660	-341.9750	-215.1640	-338.8820	0.6292	126.8110	[m s^-1]
W	346.0500	380.7610	286.4060	370.8530	0.7522	-94.3546	[m s^-1]
Distortion Parameter	1.0332	1.0936	1.0839	1.0655	0.9912	N/A	
Flow Angle: Alpha	0.0161	-2.5938	38.5660	8.3434	N/A	41.1598	[degree]
Flow Angle: Beta	-65.9902	-64.2231	-48.4508	-65.9480	N/A	15.7723	[degree]

Table 8: Component 1 summary for iteration 2

Table 3. Component 1 Summary Data Table

Quantity	Inlet	LE Cut	TE Cut	Outlet	TE/LE	TE-LE	Units
Density	1.1613	1.2421	1.2695	1.3790	1.0221	N/A	[kg m^-3]
Pstatic	94982.9000	102592.0000	116043.0000	116297.0000	1.1311	13450.9000	[Pa]
Pt	104057.0000	121912.0000	158324.0000	132903.0000	1.2987	36412.5000	[Pa]
T	284.7140	284.9560	317.2800	293.4290	1.1134	32.3233	[K]
Tt	292.3910	300.1100	348.1220	305.3450	1.1600	48.0119	[K]
H	-13495.3000	-13251.7000	19213.9000	-4742.0300	-1.4499	32465.5000	[J kg^-1]
Ht	-5784.4300	1969.0800	50192.3000	7226.2800	25.4902	48223.2000	[J kg^-1]
Ht adiabatic	-5783.5400	5579.3200	38570.4000	15311.7000	6.9131	32991.0000	[J kg^-1]
Ht polytropic	-5784.4500	5491.4600	38479.8000	15010.4000	7.0072	32988.3000	[J kg^-1]
Rothalpy	-5783.6500	2353.9200	-60.5827	-243.0550	-0.0257	-2414.5100	[J kg^-1]
Entropy	-26.9859	-38.9235	9.1678	-53.0549	-0.2355	48.0913	[J kg^-1 K^-1]
Mach (abs)	0.3619	0.4991	0.6519	0.4384	1.3062	0.1528	
Mach (rel)	1.0221	1.1257	0.7991	1.0801	0.7099	-0.3266	
U	318.1090	343.9490	355.9150	359.6400	1.0348	11.9654	[m s^-1]
Cm	122.4190	165.8090	181.7260	148.8920	1.0960	15.9172	[m s^-1]
Cu	0.0422	1.9752	140.7510	20.7577	71.2572	138.7760	[m s^-1]
C	122.4200	169.0640	233.6020	150.5600	1.3817	64.5381	[m s^-1]
Wu	-318.0660	-341.9750	-215.1640	-338.8820	0.6292	126.8110	[m s^-1]
W	346.0500	380.7610	286.4060	370.8530	0.7522	-94.3546	[m s^-1]
Distortion Parameter	1.0332	1.0936	1.0839	1.0655	0.9912	N/A	
Flow Angle: Alpha	0.0161	-2.5938	38.5660	8.3434	N/A	41.1598	[degree]
Flow Angle: Beta	-65.9902	-64.2231	-48.4508	-65.9480	N/A	15.7723	[degree]

Table 9: Component 1 summary for iteration 3

Table 3. Component 1 Summary Data Table

Quantity	Inlet	LE Cut	TE Cut	Outlet	TE/LE	TE-LE	Units
Density	1.1476	1.0324	1.0420	1.0332	1.0093	N/A	[kg m^-3]
Pstatic	92648.5000	83794.1000	95991.1000	95022.9000	1.1456	12197.0000	[Pa]
Pt	100998.0000	104564.0000	175167.0000	152737.0000	1.6752	70602.6000	[Pa]
T	280.2900	270.9400	300.8810	298.0690	1.1105	29.9409	[K]
Tt	288.1640	290.6030	350.2730	350.5620	1.2053	59.6703	[K]
H	-17939.1000	-27329.3000	2743.4200	-81.4247	-0.1004	30072.7000	[J kg^-1]
Ht	-10030.2000	-7580.6900	52352.1000	52643.1000	-6.9060	59932.8000	[J kg^-1]
Ht adiabatic	-10029.7000	-11539.0000	35105.7000	31855.6000	-3.0423	46644.7000	[J kg^-1]
Ht polytropic	-10029.8000	-11579.0000	36309.6000	33093.8000	-3.1358	47888.6000	[J kg^-1]
Rothalpy	-10037.2000	-10044.1000	-9913.5400	-9685.8500	0.9870	130.5690	[J kg^-1]
Entropy	-32.9536	-19.4147	15.5937	26.3070	-0.8032	35.0084	[J kg^-1 K^-1]
Mach (abs)	0.3441	0.5653	0.9477	0.8315	1.6766	0.3824	
Mach (rel)	1.0300	1.1222	0.5840	0.6952	0.5204	-0.5382	
U	318.1090	343.7910	355.7150	359.6390	1.0347	11.9239	[m s^-1]
Cm	115.3680	168.7840	174.9180	175.0410	1.0363	6.1341	[m s^-1]
Cu	0.1544	19.3715	266.6050	209.5130	13.7627	247.2340	[m s^-1]
C	115.4850	186.9560	344.6070	296.8260	1.8433	157.6500	[m s^-1]
Wu	-317.9540	-324.4200	-89.1107	-150.1260	0.2747	235.3100	[m s^-1]
W	346.7580	370.8070	205.4940	244.3940	0.5542	-165.3140	[m s^-1]
Distortion Parameter	1.2013	1.2015	1.1588	1.1589	0.9645	N/A	
Flow Angle: Alpha	0.1018	6.1606	-2.8022	-1.1183	N/A	-8.9628	[degree]
Flow Angle: Beta	-66.6209	-62.7854	-43.5991	-50.3532	N/A	19.1862	[degree]

The Component 1 summary data highlights significant trends in pressure, velocity, and thermodynamic quantities across the three iterations. Iterations 1 and 2 exhibit nearly identical values, with high total pressure rise but unrealistic efficiency behaviour, suggesting incomplete convergence. Iteration 3, however, shows a more balanced flow with higher total pressure and temperature gains, lower entropy increase, and better-aligned Mach number distributions. Flow angles and velocity components also shift noticeably, indicating refined flow guidance through the blades. These results align with improved performance and more accurate simulation conditions in the final iteration.

Lessons Learned

The lesson learned from this is that the more tweaks done to a system and the more iterations that are run, the more the overall system performs at optimum operations. This remained true

throughout the whole report as seen in the graphs presented. Overall, the more iterations that were run, the better the system performed.

Conclusion

This CFD assignment demonstrated a step-by-step refinement process for improving the aerodynamic performance of a turbofan engine's fan stage using ANSYS CFX and MDIDS-GT data as reference. Three design iterations were completed, with each involving geometric and boundary condition changes, including adjustments to stagger angles, leading edge diameters, and outlet pressures. Performance was evaluated based on flow stability, mass flow accuracy, efficiency, and flow separation behaviour.

The first two iterations, limited to 100 solver steps, showed signs of under-convergence, unrealistic efficiencies, and poor correlation with MDIDS targets. In contrast, the third iteration, extended to 500 steps, achieved significantly improved and physically consistent results. It produced a mass flow rate and pressure ratio closer to MDIDS values, minimized flow separation, and showed stable convergence across turbulence, energy, and momentum metrics. Mesh consistency across all cases ensured that the improvements were driven by design refinements, not numerical variation.

Overall, the results confirm that careful parameter tuning combined with adequate iteration count leads to better predictive performance in CFD. The final iteration demonstrated the closest match to design targets, validating the iterative design process and providing key insights into the importance of convergence, boundary conditions, and geometry control in turbomachinery analysis.